

Computation Graph and Backpropagation

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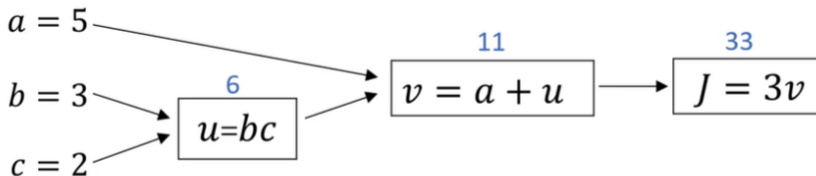
University of Management and Technology (UMT), Lahore

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Computing derivatives



Forward Recap

Target function:

$$J = 3(a + bc), \quad u = bc, \quad v = a + u, \quad J = 3v.$$

Example values: $a = 5, b = 3, c = 2 \Rightarrow u = 6, v = 11, J = 33$.

Step 1: $\frac{dJ}{dv}$

Since $J = 3v$, a small increase Δv causes $\Delta J = 3 \Delta v$.

$$\boxed{\frac{dJ}{dv} = 3}$$

Step 2: $\frac{dJ}{da}$

Because $v = a + u$, we have $\frac{dv}{da} = 1$. By the chain rule:

$$\frac{dJ}{da} = \frac{dJ}{dv} \cdot \frac{dv}{da} = 3 \times 1 = 3.$$

Step 3: $\frac{dJ}{du}$

Because $v = a + u$, we have $\frac{dv}{du} = 1$. Thus:

$$\frac{dJ}{du} = \frac{dJ}{dv} \cdot \frac{dv}{du} = 3 \times 1 = 3.$$

Step 4: $\frac{dJ}{db}$

Since $u = bc$, $\frac{du}{db} = c = 2$. Chain rule:

$$\frac{dJ}{db} = \frac{dJ}{du} \cdot \frac{du}{db} = 3 \times 2 = 6.$$

Step 5: $\frac{dJ}{dc}$

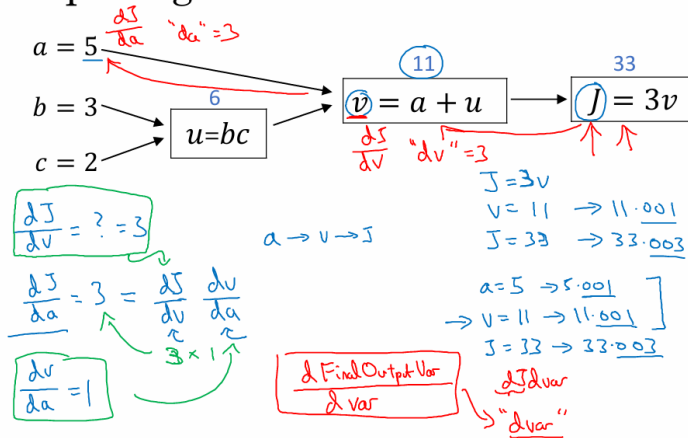
Since $u = bc$, $\frac{du}{dc} = b = 3$. Chain rule:

$$\frac{dJ}{dc} = \frac{dJ}{du} \cdot \frac{du}{dc} = 3 \times 3 = 9.$$

Summary Table

Derivative	Formula	Value
$\frac{dJ}{dv}$	3	3
$\frac{dv}{dJ}$	$\frac{dJ}{dv} \cdot 1$	3
$\frac{da}{dJ}$	$\frac{dv}{dJ} \cdot 1$	3
$\frac{du}{dJ}$	$\frac{dv}{dJ} \cdot 2$	6
$\frac{db}{dJ}$	$\frac{du}{dJ} \cdot 3$	9
$\frac{dc}{dJ}$	$\frac{du}{dJ}$	

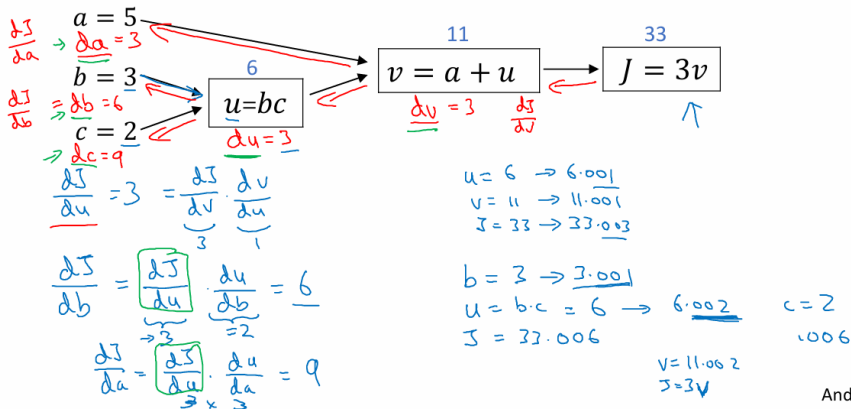
Computing derivatives



Andrew Ng

Backprop Flow on the Graph 02

Computing derivatives



Andrew Ng

Key Takeaways

- Backprop = efficient right-to-left gradient computation using the chain rule.
- Reuse intermediate gradients (e.g., $\frac{dJ}{dv}$) to compute earlier ones.
- Forms the basis for training deep neural networks.